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LiFT-NIDS: Lightweight Transformers for Network Intrusion Detection

Evaluating Lightweight Transformers for Network Intrusion Detection under Temporal Drift: A Comparative Study with MAWIFlow

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github.com/JohnGarden/sbseg26-LiFT-NIDS · v1.0.0-sbseg2026-camera-ready



SeloD

SeloF

SeloS

SeloR

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Armação dos Búzios – RJ

What This Talk Covers

A G E N D A

- 1 Why static splits fall short slides 3–6
- 2 The proposed model slide 7
- 3 Axis 1 — static sanity check slides 8–9
- 4 Axis 2 — temporal drift slides 10–15
- 5 Limitations and conclusions slides 16–17

Static Splits Measure the Wrong Thing

MOTIVATION

1

The assumption: traffic holds still

Train and test come from the same collection period, so the model is never asked to leave the distribution it learned on.

[Sommer and Paxson 2010]

2

The reality: traffic keeps changing

New attack campaigns, changing user behaviour and evolving infrastructure move the distribution — and the labels move with it.



So a near-perfect score on its own collection period says nothing about next year — and gives no warning.

The Same Four Models, Two Protocols

THE RESULT IN ONE CHART

Axis 1: Static Split (CICIoT2023)

Random 60/20/20 split. On attack-class F1, practically indistinguishable:



score on the test split — higher is better

Axis 2: Temporal Drift (MAWIFlow)

Annual forward chaining across 18 years. Dramatic separation:



score one year later — higher is better

Why This Study Is Needed

RESEARCH GAP

An Open Challenge in the Field

Systematic reviews of Transformer-based IDS name temporal adaptability, dataset dependence and scalability as open challenges; dataset surveys flag split quality and timeliness as persistent limitations.

Tested Only on Static Splits

Manocchio et al. (2024) ablated Transformer components for flow NIDS and found compact configurations competitive at far lower model size — but validated them on static benchmarks only.

No Transformer in the Benchmark

The MAWIFlow benchmark (Schraven et al., 2026) runs annual forward chaining on real backbone traffic across six model families — none of them a Transformer.

RQ1 — Does near-ceiling performance under a conventional static NIDS benchmark (CICIoT2023) constitute sufficient evidence of robustness under temporal drift?

RQ2 — Does a parameter-efficient causal Transformer retain higher temporal performance than tabular and recurrent baselines under forward-chaining evaluation?

H1 — Sequential flow representations, modeled as windows of $W \times F$ consecutive flow records, achieve higher temporal robustness (measured by $nAUT_H$) than tabular $1 \times F$ representations under temporal drift; evaluated under the cumulative training-window forward-chaining protocol.

Where This Paper Sits (Paper Table 1)

POSITIONING

Ref.	Dataset	Models	Fwd. chain	nAUT metric	Transformer	Real backbone
W1	NSL-KDD, UNSW-NB15, CSE-CIC-IDS2018	FlowTransformer (enc./dec.; GPT/BERT; ablation)	X	X	✓	X
W2	MAWIFlow (MAWILab v1.1, 2007–2024)	CNN-BiLSTM, DT, IF, LR, RF, XGBoost	✓	✓	X	✓
W3	CICIDS2017, CIC-DDoS2019	RTIDS: encoder-decoder Transformer	X	X	✓	X
W4	UNSW-NB15 (packet-level)	Decision Transformer (offline RL)	X	X	✓	X
W5	NSL-KDD	RNN-IDS	X	X	X	X
W6	ISCX VPN-nonVPN	End-to-end 1D-CNN	X	X	X	X
This paper	MAWIFlow + CICIOT2023	XGBoost, MLP, CNN-BiLSTM, LightTransformer	✓	✓	✓	✓

Notes: ✓ = present; X = absent or not applicable. W1 [Manocchio et al. 2024]; W2 [Schraven et al. 2026]; W3 [Wu et al. 2022]; W4 [Zhou et al. 2024]; W5 [Yin et al. 2017]; W6 [Wang et al. 2017].

Two gaps meet here: the compact FlowTransformer configuration (W1) was validated on static benchmarks only, and the MAWIFlow forward-chaining benchmark (W2) carries no Transformer baseline — this paper is the first row that ticks all four columns.

LightTransformer Architecture

PROPOSED CONFIGURATION

Flow window

16 consecutive flows × 89 features

16 × 89



Record-level projection + sinusoidal PE

linear per record, then positional encoding

16 × 128



Causal decoder block × 2

2 heads · FF 512 · Pre-LN · causal mask

16 × 128



Last-token readout

take position $t = T$ only

128



MLP classification head

128 → 64 → 2 logits

2

Adopted, not invented

The compact configuration from the FlowTransformer ablation of Manocchio et al. (2024), reimplemented in PyTorch.

416k

parameters

148 MB

peak VRAM

~29 μ s

inference / sample

8 GB

laptop GPU

Axis 1: Static Split on CICIoT2023

SANITY CHECK

Model	F1 ⁺ (attack class)	Macro F1	AUC-ROC
XGBoost	0.9967 [0.9967, 0.9967]	0.9281	0.9985
MLP	0.9913 [0.9913, 0.9914]	0.8638	0.9976
LightTransformer	0.9894 [0.9888, 0.9903]	0.8413	0.9949
CNN-BiLSTM	0.9892 [0.9889, 0.9895]	0.8404	0.9956

Purpose of Axis 1: 60/20/20 stratified split, R = 5 seeds. Attack is 97.7% of CICIoT2023, so F1⁺ is near-trivially high and macro F1 is the primary ranking metric. CICIoT2023 carries no timestamp, so the sequential windows hold no temporal order — read this axis as static capacity only.

1

Everything clears the ceiling

All four families reach $F1^+ > 0.98$ on CICIoT2023 — where attack is 97.7% of the data.

2

Macro F1 does rank them

It separates the families and puts XGBoost first at 0.928, six to nine points above the neural models.



And XGBoost ranks third under drift — a static ranking is not a deployment ranking.

MAWIFlow-subset: 18 Years of Backbone Traffic

D A T A

TIMELINE

18 Years

2007 – 2024

CAPTURE

74 Days

1 day per quarter, where published

TOTAL FLOWS

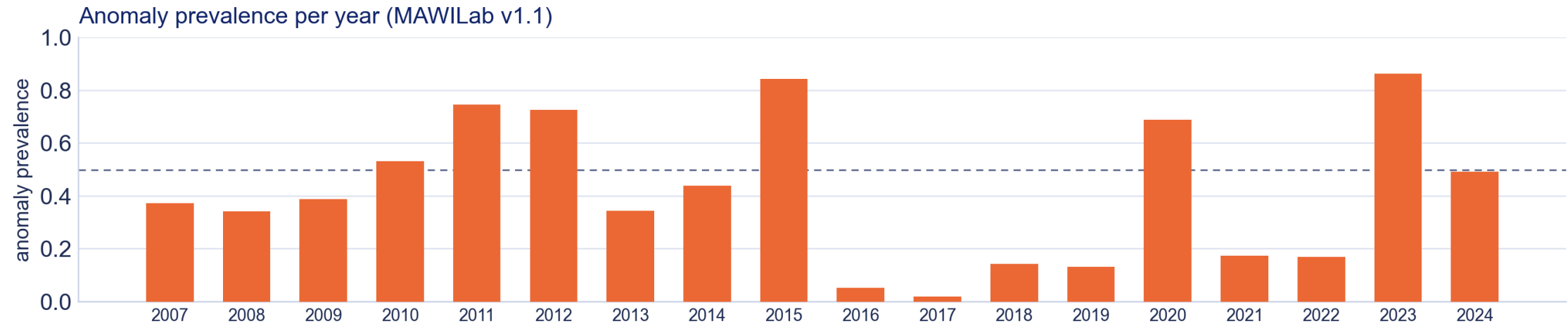
9.95M

9,946,112 flows

FEATURES

89

numeric features



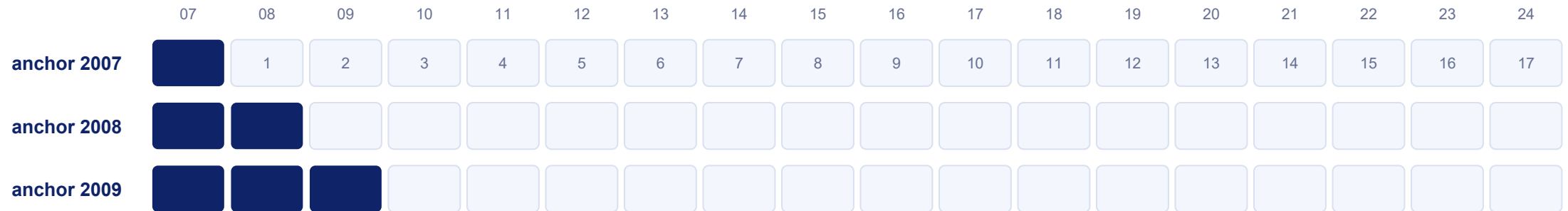
2.1% (2017) to 86.3% (2023) — the drift is in the label distribution, not only in the features.

Volume is uneven across the span: most post-2013 years hold 7k–52k flows, against 2.7M in 2007 and 971k in 2023.

Axis 2: Annual Forward Chaining

PROTOCOL

Phase 2 — train on past years, test on every later year, slide the anchor forward



... one window per anchor year, 2007–2023 — 17 windows



training window



test years, evaluated one by one

numbers = Δt , the gap from the anchor

Phase 1 — tune once, then freeze

Optuna HPO runs on the 2007 anchor year alone, on a temporal 60/20/20 split.
No future year touches hyperparameters, preprocessing or early stopping.

Phase 3 — aggregate by temporal gap

Median $F1^+$ at each Δt , trapezoidal integration per seed, averaged over $R = 5$ seeds $\rightarrow nAUT_1, nAUT_3, nAUT_5$.

Four training windows: $k = 1, 2, 3$ and cumulative (drawn above). 16 combinations $\times$ 5 seeds = 80 runs, all completed, 0 run failures.

nAUT: One Number for a Degradation Curve

METRIC

$$\text{nAUT}_H = \frac{1}{H} \sum_{\Delta t=0}^{H-1} \frac{\tilde{F}_1^+(\Delta t) + \tilde{F}_1^+(\Delta t + 1)}{2}$$

$\tilde{F}_1^+(\Delta t)$: the median positive-class F1 (MAWILab anomaly) at a temporal gap of Δt years, taken over every training window that yields that lag.

$\Delta t = 0$: the held-out 20% slice of the training window — in-distribution, but never used to fit weights.

Normalisation: trapezoidal integration per seed, normalised by H , then averaged over $R = 5$. nAUT stays in $[0, 1]$, comparable across horizons. Higher is better.

Metric definition from the MAWIFlow benchmark (Schraven et al., 2026), unchanged.

Axis 2: A Different Ordering Under Drift

RESULTS

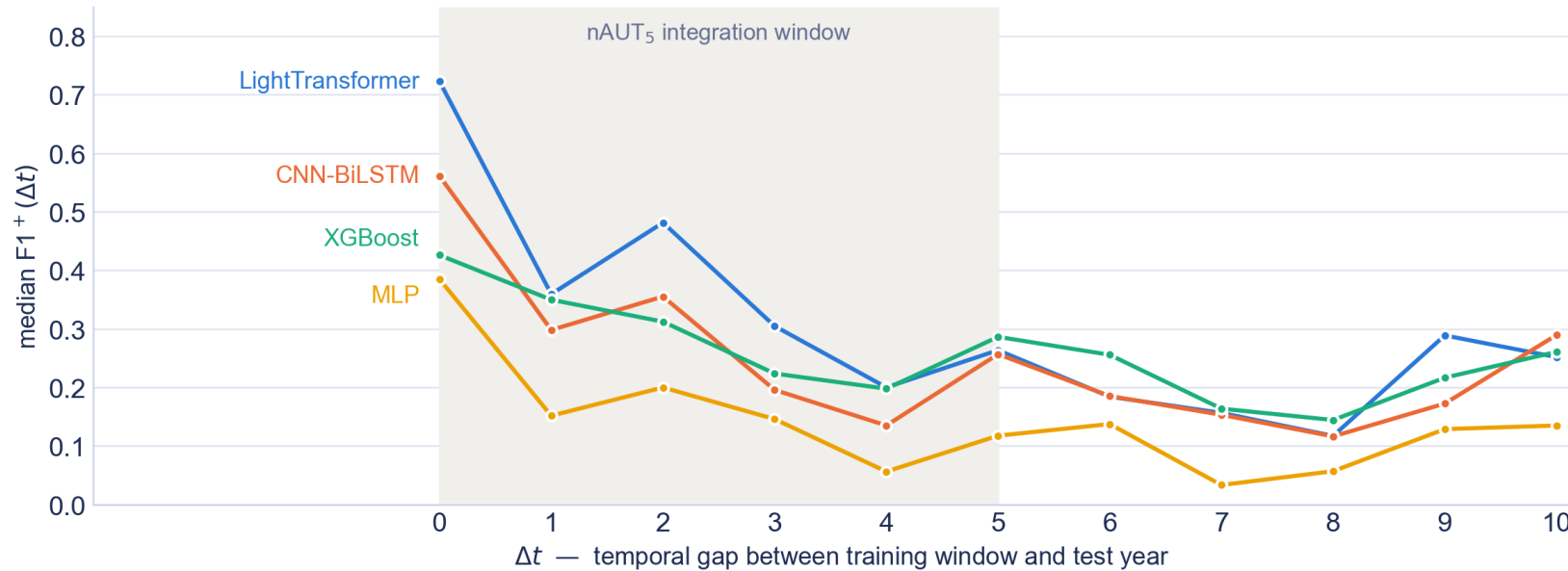
Model	nAUT ₁ (H = 1 year)	nAUT ₃ (H = 3 years)	nAUT ₅ (H = 5 years)
LightTransformer	0.568 [0.519, 0.618]	0.474 [0.437, 0.503]	0.386 [0.360, 0.409]
CNN-BiLSTM	0.439 [0.418, 0.463]	0.358 [0.338, 0.377]	0.289 [0.273, 0.303]
XGBoost	0.377 [0.350, 0.396]	0.325 [0.308, 0.337]	0.285 [0.269, 0.297]
MLP	0.282 [0.260, 0.303]	0.210 [0.203, 0.221]	0.165 [0.158, 0.173]

Headline: cumulative training window. The LightTransformer leads at every horizon and its CI never overlaps CNN-BiLSTM's, though the absolute gap narrows with horizon — 0.129 at H = 1, 0.115 at H = 3, 0.097 at H = 5. Interval overlap is not a formal paired test — but the LightTransformer's worst seed still beats CNN-BiLSTM's best.

Paper Table 6: means over R = 5 seeds, brackets are bootstrap 95% confidence intervals.

What the Degradation Actually Looks Like

TRAJECTORY



$\Delta t = 0$ is the held-out 20% slice of the training window (in-distribution), not the training data itself.

Seeds carrying trajectory data: XGBoost 5, LightTransformer 4, CNN-BiLSTM 4, MLP 2 - the scalar nAUT values are complete for 5/5 seeds on all four models.

One year of gap halves the in-distribution $F1^+$ of the three neural models (47–61%), but costs XGBoost only 18% — from a far lower start.

The Advantage Depends on Training Memory

S E N S I T I V I T Y

Training Window	LightTransformer	CNN-BiLSTM	XGBoost	MLP
k = 1 year (short memory)	0.30 (last)	0.32	0.39 (best)	0.37
k = 2 years	0.36 (best)	0.34	0.32	0.26
k = 3 years	0.37 (best)	0.27	0.19	0.20
Cumulative window (full history)	0.57 (best)	0.44	0.38	0.28

Architectural reading: cells are mean $nAUT_1$ over $R = 5$ seeds. The Transformer's advantage needs a long history: at $k = 1$ it is the worst model, from $k = 2$ it leads, widest on the cumulative window. These fixed-window results are descriptive only — no inferential tests were pre-specified, and every comparative conclusion is scoped to the cumulative window.

Sample size ($n = 5$ seeds)

No pairwise comparison survives strict Bonferroni correction — including $LT > CNN-BiLSTM$, the study's primary contrast. The other five pairs are exploratory.

MAWIFlow-subset, one backbone

One capture day per quarter where MAWILab v1.1 publishes one (74 days / 18 years) from one trans-Pacific backbone link. Most post-2013 years hold 7k–52k flows. Not a MAWIFlow benchmark reproduction.

Unequal HPO budgets

50 Optuna trials for XGBoost against 1–15 for the neural models on Axis 1 (GPU-bound). This favours the tabular baselines — it cuts against our own hypothesis.

Internal validity — no isolating ablation

Representation, architecture, capacity and training dynamics vary together; the isolating ablation (LT at $W = 1$) was not executed, so sequential context cannot be credited causally.

RQ1, RQ2, H1 — and the Recommendation

C O N C L U S I O N S

RQ1 — addressed

Near-ceiling static scores are not evidence of temporal robustness: $F1^+ > 0.98$ for all four families on C10T2023 coexists with a twofold $nAUT_1$ spread, 0.28 to 0.57, on MAWIFlow. Evidence on slides 8–9 and 13.

RQ2 — answered

Under cumulative forward chaining the LightTransformer holds the highest $nAUT$ at every horizon (0.568 / 0.474 / 0.386), in all five seeds; on $nAUT_1$, Friedman $\chi^2 = 15.0$, $p = 0.002$. Evidence on slide 13.

H1 — empirically supported, scoped

Both $W \times F$ sequential models beat both $1 \times F$ tabular baselines on $nAUT_1$, in all five seeds — under the cumulative training window, the only regime the paper draws comparative conclusions from. Evidence on slides 13 and 15.

What we do not claim

No causal attribution to sequential context: the isolating ablation (LT at $W = 1$) was not executed. No corrected pairwise significance — nothing survives Bonferroni at $\alpha = 0.0083$. No invariance to the training-window regime.

Temporal forward-chaining evaluation belongs in deployment-oriented NIDS benchmarking — alongside the static split, not instead of it.

Thank you for your attention

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github.com/JohnGarden/sbseg26-LiFT-NIDS

release v1.0.0-sbseg2026-camera-ready

